

NETWORKED CONTROL

Prof. Marcello Farina

NAME/FAMILY NAME..... SOLUTIONS.....
PERSONAL CODE.....
SIGNATURE.....
DATE..... 23/1/'23.....

Notes:

- The solutions must be written in the pages of this leaflet.
- At the end of the exam only this file is accepted, **without any additional page**.
- No notes or books are accepted, neither in paper nor electronic form.
- Clarity and precision will be matter of evaluation.

A. a) $\lambda = 1 > 0$ THE SYSTEM IS UNSTABLE
 MODE: e^t

b) $u = Kx \quad \dot{x} = (1 + 2K)x$

FOR STABILITY: $1 + 2K < 0 \quad K < -\frac{1}{2}$

B. $x_{k+1} = Fx_k + Gu_k \quad F = e^h = e^{\log 2} = 2$
 $y_k = Hx_k \quad G = \frac{1}{a} (e^{ah} - 1) = 2$
 $H = 1$

$\Rightarrow x_{k+1} = 2x_k + 2u_k$

$u_k = Ky_k = Kx_k \quad \Rightarrow x_{k+1} = (2 + 2K)x_k$

FOR STABILITY: $|2K + 2| < 1 \quad \Leftrightarrow \begin{cases} K < -\frac{1}{2} \\ K > -\frac{3}{2} \end{cases}$

$\Rightarrow -\frac{3}{2} < K < -\frac{1}{2}$

c. a. $h = \log 2, k = -1$

$\Phi(h, z) = \begin{bmatrix} e^h + \Gamma(h-z)zk & e^{(h-z)}\Gamma(z)zk \\ 1 & 0 \end{bmatrix}$

$e^{-z} = \alpha$

$e^h = 2$

$e^{h-z} = e^{\log 2} e^{-z} = 2\alpha = 2\alpha$

$\Gamma(h-z) = (e^{h-z} - 1) = 2\alpha - 1$

$\Gamma(z) = e^z - 1$

$\Rightarrow \Phi(h, z) = \begin{bmatrix} 2 - 2(2\alpha - 1) & -4e^{-z}(e^z - 1) \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 4 - 4\alpha & 4(\alpha - 1) \\ 1 & 0 \end{bmatrix}$

THE CHARACTERISTIC POLYNOMIAL OF $\Phi(z)$ IS

$$p(\lambda) = \det(\lambda I - \Phi(z)) = \det \begin{bmatrix} \lambda - 4(1-\alpha) & -4(\alpha-1) \\ -1 & \lambda \end{bmatrix}$$

$$= \lambda^2 - (4 - 4\alpha)\lambda - 4(\alpha-1)$$

CONDITION FOR SCHUR STABILITY:

$$(-1 < 4(\alpha-1) < 1) \wedge (-1 + 4(\alpha-1) < 4\alpha - 4 < +1 - 4(\alpha-1))$$

$$4\alpha - 4 < 1 \Leftrightarrow \alpha < \frac{5}{4}$$

$$4\alpha - 4 > -1 \Leftrightarrow \alpha > \frac{3}{4}$$

$$4\alpha - 4 > -1 - 4 + 4\alpha \Rightarrow -1 < 0 \quad \checkmark$$

$$4\alpha - 4 < 5 - 4\alpha \Leftrightarrow 8\alpha < 9 \Leftrightarrow \alpha < \frac{9}{8}$$

$$\Leftrightarrow \begin{cases} e^{-z} < \frac{9}{8} \\ e^{-z} > \frac{3}{4} \end{cases} \quad \forall z > 0$$

$\Uparrow$

$$-z > \log\left(\frac{3}{4}\right)$$

$\Uparrow$

$$z < \left| \log_e\left(\frac{3}{4}\right) \right| = \bar{z}$$

b. $z=0$, $\theta_k = 0, 1, 0, 1, \dots$

$$\hat{\Phi}_1 = \begin{bmatrix} F + GK & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 + 2K & 0 \\ 1 & 0 \end{bmatrix}$$

$$\hat{\Phi}_0 = \begin{bmatrix} F & GK \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2K \\ 0 & 1 \end{bmatrix}$$

THE 2-STEPS TRANSITION MATRIX IS, E.G.

$$\hat{\Phi}_1 \hat{\Phi}_0 = \begin{bmatrix} 2 & 2K \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 + 2K & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 4 + 6K & 0 \\ 1 & 0 \end{bmatrix}$$

THE EIGENVALUES ARE $\lambda_1 = 4 + 6K$ AND $\lambda_2 = 0$

THE NCS IS AS. STABLE IFF $|6K + 4| < 1$ I.E.:

$$\begin{cases} 6K + 4 < 1 \\ 6K + 4 > -1 \end{cases} \Leftrightarrow -\frac{5}{6} < K < -\frac{1}{2}$$

